

Elaine Zhu

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TECHNICAL SKILLS

Languages: Python, C, C++, C#, HTML, CSS, JavaScript/TypeScript, Verilog

Tools and Technologies: React, MySQL, Figma, Linux, Supabase, Unity, Godot, Git, Blender

EXPERIENCE

Engineering Intern | Code Ninjas

May 2026 – Aug 2026

- Built an AI-powered report generation tool using **React, TypeScript, and Gemini API**, automating personalized student progress reports and reducing report generation time by **80%+**
- Instructed **100+** students (ages 4–16) in weekly coding lessons using Unity (C#), Godot (GDScript), JavaScript, Roblox Studio (Lua), and web technologies, building programming skills and confidence
- Designed studio-wide initiatives, including Project of the Week and coding challenges, to boost student morale

Software Engineering Intern | Mespere LifeSciences

Aug 2024 – Jan 2025

- Developed a remote checkup platform that improved patient accessibility using **React, TypeScript, and Next.js**, deployed in a **\$2M** clinical research project
- Built tools in React to visualize systolic and diastolic blood pressure trends across 3-month to 1-year periods, enabling fast and accurate interpretation for doctors
- Implemented analytics and reporting workflow to support clinical insights and structured summaries

PROJECTS

Nine Levels Down | Godot, GDScript, Blender

July 2026

- Built the minigame-switching system (LevelManager, MinigameBase), enabling dynamic, drop-in minigames
- Implemented first-person shooting via PhysicsRayQueryParameters3D raycasting against Area3D collision groups

WatCard Dashboard | React, TypeScript, Python, Node.js/Express

Oct 2025 – Dec 2025

- Led front-end development for a student card budget dashboard using **React and TypeScript**, integrating data from a custom Chrome extension and **Node.js/Express** API
- Drove sprint planning and task prioritization while implementing unit tests and coordinating feature delivery through GitLab issue boards

AWARDS AND LEADERSHIP

CFUW Dr. Mabel Dunham Award

July 2025

- Highest university entry average (99.2%) for female graduates at LHSS

Volunteer Lead | Hack the North

May 2026 – Present

- Served as Volunteer Food Lead, coordinating fast, organized food delegation for **1000+** attendees
- Organized a training event and created onboarding materials for a team of **600+** volunteers

Marketing Co-Director | Waterloo Engineering Competition/EngHacks

Oct 2025 – Present

- Partnered with the technical team to design the websites in Figma, for events with ≈ 400 participants
- Designed promotional materials and managed social media across **3 terms**, reaching **60k+** viewers in a week

Marketing Co-Lead | SVP Teens

Aug 2023 – July 2025

- Led youth volunteer initiatives contributing **200+ volunteer hours** and **raising \$8,000+** for local charities
- Organized fundraising events (**100+** attendees) and designed promotional content reaching **30k+** viewers

EDUCATION

Bachelor of Software Engineering, University of Waterloo

Sept 2025 – Present

- Awarded President's Scholarship of Distinction GPA 4.0
- Relevant Coursework:** Programming Principles, Data Abstraction & Implementation, Digital Circuits & Systems, Intro Methods to Software Engineering, Linear Algebra for Engineering, Public Speaking